

The schematic diagram illustrates the power supply and USB interface for the AP2112K-3.3 module. The module is powered by a 3.3V supply, and the USB interface is connected to a USB-C port. The diagram shows the connection of a USB-C connector (J1) to the module's pins, including VBUS, CC1, CC2, USB_DM, and USB_DP. It also shows the internal USB controller (U4) and the power regulation section with capacitors C1 through C14. The module is powered by a 3.3V supply, and the USB interface is connected to a USB-C port.

USB nach seriell, Auto-Reset

Rechner

Pegelwandlung RS232

Anschlüsse

The diagram illustrates the hardware setup for an ESP32-WROOM-32E module. It is divided into four main sections:

- USB nach seriell, Auto-Reset:** This section shows the USB-to-serial converter circuit. It includes a USB-to-serial chip (CH340K) connected to a USB DP/DM pair (pins 1, 2) and a 3V3 supply. The chip's TXD, RXD, DTR, and RTS pins are connected to the ESP32's TXD0, RXD0, DTR, and RTS pins. A 100nF capacitor (C5) is connected to the 3V3 supply. The ESP32's EN pin is connected to a reset switch (SW1) and a boot switch (SW2).
- Rechner:** This section shows the connection to a computer. The ESP32's TXD0, RXD0, DTR, and RTS pins are connected to the computer's TXD, RXD, DTR, and RTS pins. The ESP32's EN pin is connected to the computer's EN pin.
- Pegelwandlung RS232:** This section shows the RS232 level shifter circuit. It includes an RS232 level shifter chip (MAX3232) connected to a 3V3 supply. The chip's TXD, RXD, DTR, and RTS pins are connected to the ESP32's TXD0, RXD0, DTR, and RTS pins. The chip's TXD, RXD, DTR, and RTS pins are connected to the computer's TXD, RXD, DTR, and RTS pins.
- Anschlüsse:** This section shows the component pin connections. The ESP32's TXD0, RXD0, DTR, and RTS pins are connected to the computer's TXD, RXD, DTR, and RTS pins. The ESP32's EN pin is connected to the computer's EN pin. The ESP32's TXD0, RXD0, DTR, and RTS pins are connected to the computer's TXD, RXD, DTR, and RTS pins.

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The schematic diagram illustrates the connections for the ESP32-WROOM-32E module. Key components and their connections are as follows:

- Power Section:** A 3.3V voltage divider (D1) is connected to the module's VDD pin. Decoupling capacitors C8 (100nF) and C9 (100nF) are connected to GND and VDD.
- Input/Output Section:** A push button (SW3) labeled 'TASTER' is connected to the module's GND and a pin. An LED (LED_D) with a 330 ohm resistor (R7) is connected to the module's GND and a pin.
- Communication Section:** Headers JP3, JP4, JP1, and JP2 are shown, connecting the module's I2C and UART pins to external components. JP3 and JP4 are labeled 'CC-RXD', 'CC-TXD', 'MAX_R2IN', 'RX-Wahl', 'MAX_T2OUT', and 'TX-Wahl'. JP1 and JP2 are labeled 'KWL-RXD', 'KWL-TXD', 'RX-Wahl', and 'MAX_R1IN'.
- Other Components:** A DS18B20 temperature sensor is connected to the module's GND and a pin. A J3 header is labeled 'CC-EASE'.